The size rule has the lowest error a practitioner can get on 10 of 15 claims

lowest a practitioner can get | tied with it | lowest of all seven | tied with it

tied: the 95% interval of the paired difference reaches zero

(a) Error of each option

(b) Change per case

	options a practitioner can apply				options that require market shares			
	KDE uniform	Log-normal uniform	Normal uniform	size rule (uniform)	KDE market	Log-normal market	Normal market	median
what is the building total?								
the total: its standard deviation	24.2	28.0	34.3	23.7	24.4	29.4	35.0	10
the total: its mean	9.8	9.9	13.5	9.6	10.0	9.4	14.2	6
the total: its 90th percentile	13.0	13.2	13.3	12.7	12.6	12.7	13.4	5
the chance of meeting a budget	6.2	5.5	10.1	5.2	6.2	5.3	9.6	3
which material is biggest, and how often?								
a material: its chance of being largest	33.5	32.4	48.8	32.6	33.3	33.7	50.5	24
a material: its share at the building 95th	29.5	27.5	28.3	27.2	27.5	26.0	26.7	10
a material: its standard deviation	30.8	32.8	36.0	30.7	29.5	33.6	36.4	9
a material: its share of the total	12.9	12.4	16.3	12.5	12.5	12.2	16.7	7
a material: its 95th percentile	24.0	23.3	24.1	22.6	21.9	21.5	22.4	6
a material: its mean contribution	16.1	15.5	20.9	15.6	14.8	13.8	21.2	5
which material drives the uncertainty?								
the uncertainty index	49.1	49.8	48.6	48.8	47.6	49.6	47.7	15
how effective is a reduction strategy?								
a cap: how often it binds	32.5	32.9	57.4	31.8	31.1	33.3	58.4	26
a cap: its chance of saving 5%	34.5	34.7	61.1	33.6	32.5	34.3	61.3	25
a cap: its mean saving	40.1	39.9	55.2	39.5	37.2	37.0	56.5	17
is this design better than that one?								
the probability B beats A	13.6	13.2	19.4	13.3	12.6	13.4	19.6	8

0 40 80

% of the mean true value

(c) The lowest error changes with dataset size

	KDE uniform	Log-normal uniform	Normal uniform	KDE market	Log-normal market	Normal market
3-9 EPDs	43.9	43.1	45.9	47.1	49.0	49.3
10-99 EPDs	29.3	27.1	31.9	30.2	28.7	32.3
100-999 EPDs	20.2	20.5	25.5	17.1	16.7	23.6
1,000+ EPDs	18.5	20.0	24.3	12.5	14.5	21.5

0 10 20 30 40 50 60

ratio of means, % of the mean true value of each claim